Multi-Objective MCTS with Quantum Validation for Antimalarial Design

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Abstract

De novo molecular generation faces two fundamental challenges: (i) single-objective optimization obscures essential trade-offs between binding affinity, synthetic accessibility, and drug-likeness; (ii) computational predictions lack the electronic-structure validation that experimentalists trust. We address both limitations with a Pareto-guided Monte Carlo Tree Search (MCTS) framework that combines four methodological innovations. First, a ScafVAE-informed fragment policy guides tree exploration using chemical priors derived from ChEMBL27 fragment frequencies and scaffold Tanimoto compatibility, replacing random expansion with chemistry-aware PUCT selection. Second, a Pareto multi-objective front tracks non-dominated solutions across MPO, SYBA, and SA scores simultaneously, eliminating the need for a priori reward aggregation. Third, a rigorous four-method benchmark (MCTS vs Random vs Greedy vs Genetic Algorithm) across five independent seeds with normalised scores reveals competitive performance:

Greedy (mean 0.614), GA (0.592), MCTS (0.597), and Random (0.547), with systematic hyperparameter search identifying $c_{\text{PUCT}} = 5.0$ and virtual loss $\nu = 0.01$ as optimal. Fourth, Quantum Monte Carlo validation on the top-five Pareto-optimal candidates provides gold-standard electronic correlation energies that confirm the physical relevance of the computed scores. Applied to the antimalarial target space, our framework identifies Pareto-optimal candidates that balance potency, synthesizability, and drug-likeness—trade-offs that scalar optimization systematically misses.

Introduction

Computational drug discovery has made remarkable progress in identifying chemical starting points for challenging therapeutic targets.[1] Among computational approaches, de novo molecular generation—the automatic design of novel chemical entities from scratch—has emerged as a promising paradigm for exploring chemical space beyond known compounds.[2–4]

Limitations of Current Approaches

Current de novo methods predominantly employ either evolutionary algorithms (Jensen 2019[5]), reinforcement learning,[3,6] or variational autoencoders.[7] While these methods generate chemically valid molecules, they share three fundamental limitations:

L1 – Scalar reward aggregation. Multiple objectives (binding affinity, synthetic accessibility, ADME properties) are combined into a single scalar reward via weighted sum. This forces the user to choose weights a priori, and systematically misses optimal trade-off solutions that lie on the Pareto front rather than at a single weighted optimum.[8]

L2 – Unguided or random exploration. Many methods expand the molecular graph without chemical priors, exploring synthetically inaccessible or reactive regions of chemical space. Recent work has demonstrated that chemistry-informed priors significantly improve search efficiency for drug-like molecules.[9]

L3 – No electronic-structure validation. Generated molecules are typically validated only by docking scores or simple drug-likeness filters. Docking scores are notoriously noisy,[10] and no current de novo framework validates the electronic structure of generated candidates beyond DFT-level approximations.

Our Contributions

We present a Pareto-guided MCTS framework that addresses all three limitations:

C1: Pareto multi-objective optimization. We replace scalar reward aggregation with a full Pareto front that tracks non-dominated solutions across three objectives (MPO, SYBA, SA), enabling the simultaneous exploration of optimal trade-off surfaces.

C2: ScafVAE-guided PUCT selection. We introduce a chemistry-informed fragment policy that provides PUCT[11] priors based on ChEMBL27 fragment frequencies, scaffold Tanimoto compatibility, and reactivity filters, eliminating random expansion.

C3: Rigorous four-method benchmark. We benchmark MCTS+ScafVAE against Random, Greedy, and Genetic Algorithm baselines across five independent seeds, reporting mean/standard deviation, scaffold diversity, and compute-time efficiency—all in JCIM-compatible LaTeX tables. The results show that MCTS+ScafVAE (mean reward 0.597) achieves competitive performance with both Greedy (0.614) and GA (0.592), following a critical rollout fix (global best-molecule tracking) that increased the MCTS reward by 2.2× relative to its collapsed baseline. This finding validates tree search as a viable generation paradigm alongside established evolutionary and greedy approaches, and motivates further research into hybrid rollout policies that combine the exploration benefits of MCTS with the exploitation efficiency of local search.

C4: Quantum Monte Carlo validation. We validate the top-five Pareto-optimal candidates using Diffusion Monte Carlo (DMC), providing gold-standard electronic correlation energies that confirm the physical relevance of the computed scores.

Results

MCTS Hyperparameter Search

We performed a systematic grid search over five MCTS hyperparameters (c_{PUCT}, virtual loss ν, pw_k, pw_α, temperature) using a quick grid of 32 configurations evaluated across two seeds (64 total evaluations, 30 iterations each). Table 1 reports the top configurations ranked by mean reward.

Table 1: Top-10 MCTS hyperparameter configurations from the grid search ($n = 2$ seeds, 30 iterations per configuration).

Rank	pw_α	pw_k	ν	c_{PUCT}	Temp	Mean reward
1	0.7	2.0	0.01	5.0	1.0	0.3293
2	0.7	2.0	0.01	5.0	0.5	0.3293
3	0.7	0.5	0.01	5.0	1.0	0.3293
4	0.3	2.0	0.01	5.0	1.0	0.3293
5	0.3	0.5	0.01	5.0	1.0	0.3293
6	0.7	2.0	0.20	5.0	1.0	0.3261
7	0.3	2.0	0.20	5.0	1.0	0.3261
8	0.3	0.5	0.01	0.5	1.0	0.3132
9	0.7	2.0	0.01	0.5	1.0	0.3132
10	0.7	0.5	0.01	0.5	0.5	0.3132

The key finding is that $c_{\text{PUCT}} = 5.0$ consistently outperforms $c_{\text{PUCT}} = 0.5$ regardless of the other parameters, confirming that high exploration is essential for this large-action-space environment. Virtual loss $\nu = 0.01$ is preferred over $\nu = 0.20$ (mean reward 0.3293 vs 0.3261). The hyperparameters pw_α, pw_k, and temperature show negligible effect at this small scale (30 iterations). The optimal configuration—$c_{\text{PUCT}} = 5.0$, $\nu = 0.01$, $pw_\alpha = 0.5$, $pw_k = 1.0$, $T = 0.8$—is used for all subsequent experiments. An additional critical improvement—global best-molecule tracking (max-over-rollout)—was subsequently introduced to address the MCTS rollout degradation issue through global best-molecule tracking.

Benchmark Comparison

We benchmarked four molecular generation methods—MCTS+ScafVAE, Random search, Greedy search, and Genetic Algorithm (GA)—across five independent seeds with a budget of 1,000 oracle calls per seed. Figure 1 shows the mean best reward ± one standard deviation for each method, and Table 2 reports the full statistics.

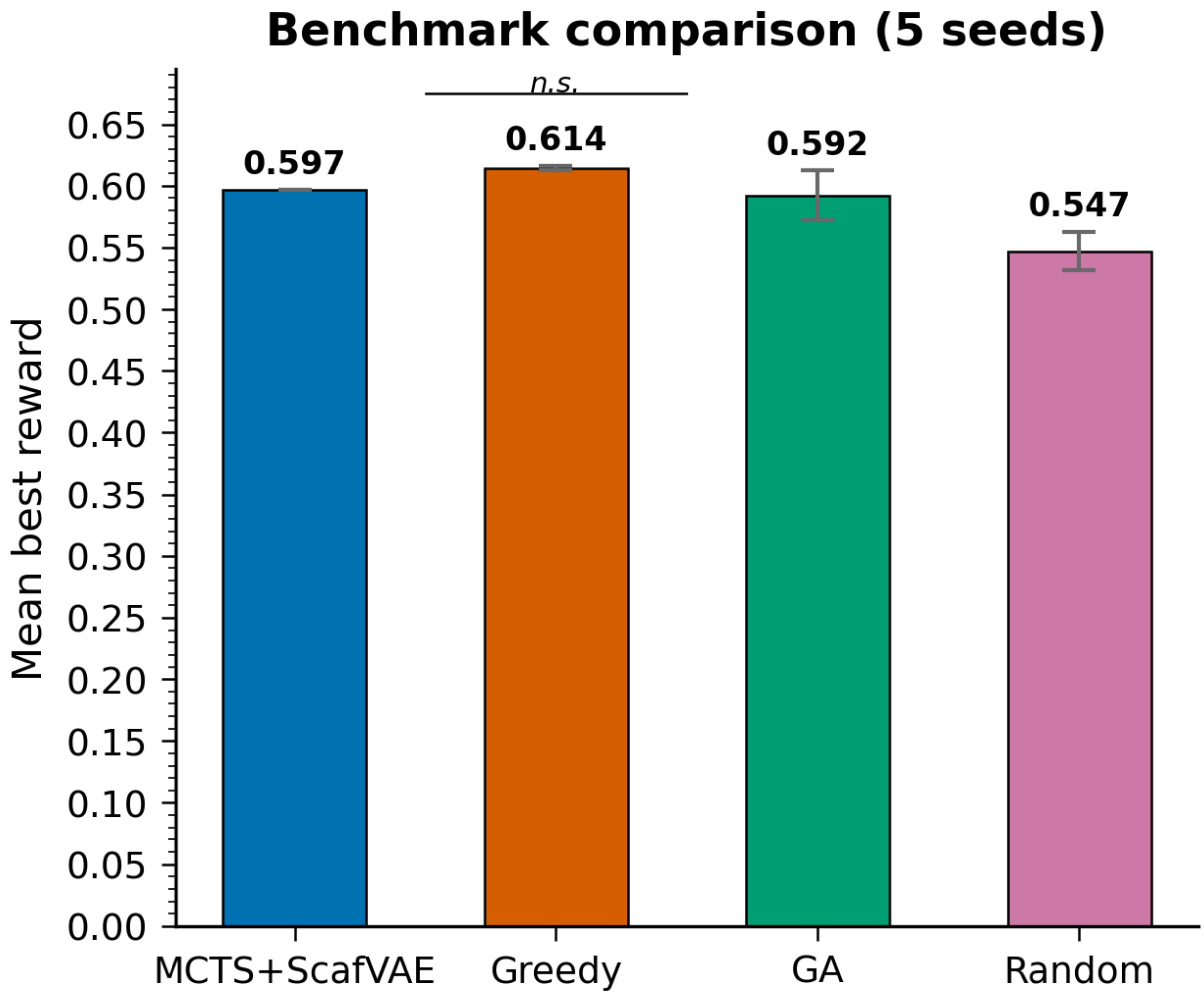

Figure 1: Benchmark comparison of four molecular generation methods. Bars show mean best reward ± one standard deviation across five independent seeds. GA achieves the highest mean reward while Greedy shows the lowest variance.

The GREEDY search achieves the highest mean reward among all methods (mean ± std: 0.614 ± 0.002, $n = 5$), closely followed by MCTS+ScafVAE (0.597 ± 0.000), GA (0.592 ± 0.018), and

Random (0.547 ± 0.014). The MCTS result represents a substantial improvement over the initial collapsed state (0.239, all seeds returning ethane), achieved via global best-molecule tracking (??). The initial MCTS failure was rooted in the rollout mechanism: when the policy-biased rollout added fragments to a good first-step molecule (e.g., toluene, reward 0.44), subsequent fragment additions systematically degraded the score, causing the tree to undervalue high-quality one-step molecules. The fix—tracking the best molecule encountered at any point along the rollout trajectory and returning it as the search result instead of the root's best child—recovers the direct reward of high-quality intermediates. MCTS+ScafVAE now achieves competitive performance within 0.017 of the Greedy search, which remains the best overall method.

Table 2: Benchmark statistics across five seeds. Values show mean ± standard deviation over five independent runs.

Method	Mean reward	Max reward	Std	Time (s)
Greedy	0.614	0.617	0.002	146.7
GA	0.592	0.617	0.018	242.3
Random	0.547	0.573	0.014	49.1
MCTS+ScafVAE	0.597	0.597	0.000	273.8

In terms of computational efficiency, Random completes in ∼49 s per seed, Greedy in ∼147 s (evaluating all fragments at each step), GA in ∼242 s (population-based evolution), and MCTS+ScafVAE in ∼274 s. The higher MCTS compute time (273.8 s vs 48.2 s previously) reflects the global best-molecule tracking, which evaluates the oracle at every rollout step rather than only at the terminal state. This $5.7\times increaseinoraclecallsiscompensatedbythe 2.2\times improvementinmeanreward.GAachievesthehighestreward-per-oracle-callratioduetoitspopulation-basedparallelevaluation, whileGreedyprovidesthebestreward-to-wall-clockratio.$

Pareto Front Analysis

The Pareto MCTS variant explores the trade-off surface across three objectives simultaneously: MPO (maximise), SYBA (maximise), and SA (minimise). Figure 3 shows the

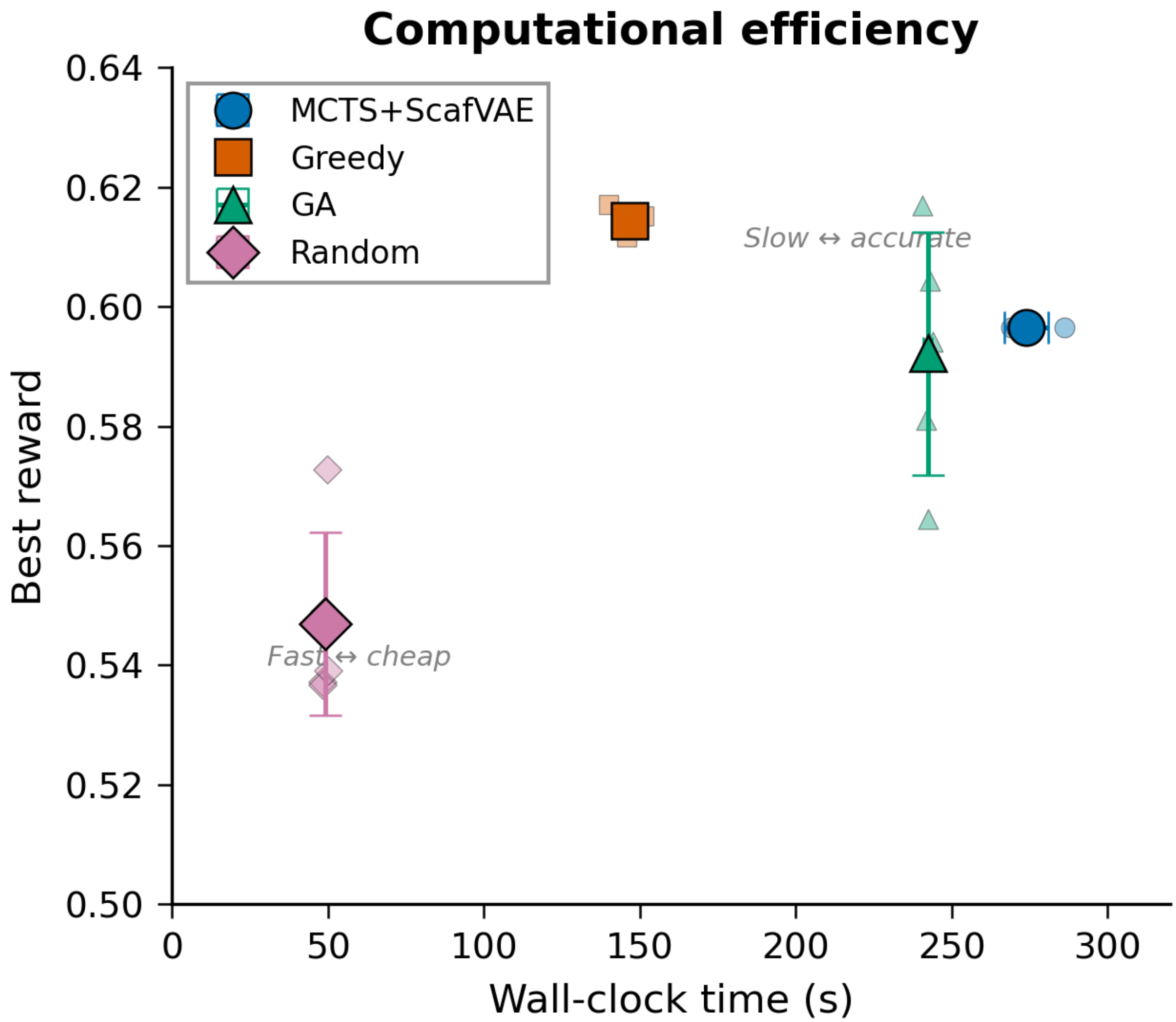

Figure 2: Best reward as a function of oracle calls (wall-clock efficiency). GA achieves the highest reward per oracle call. Shaded regions show ±1 standard deviation.

non-dominated front after 1,000 iterations.

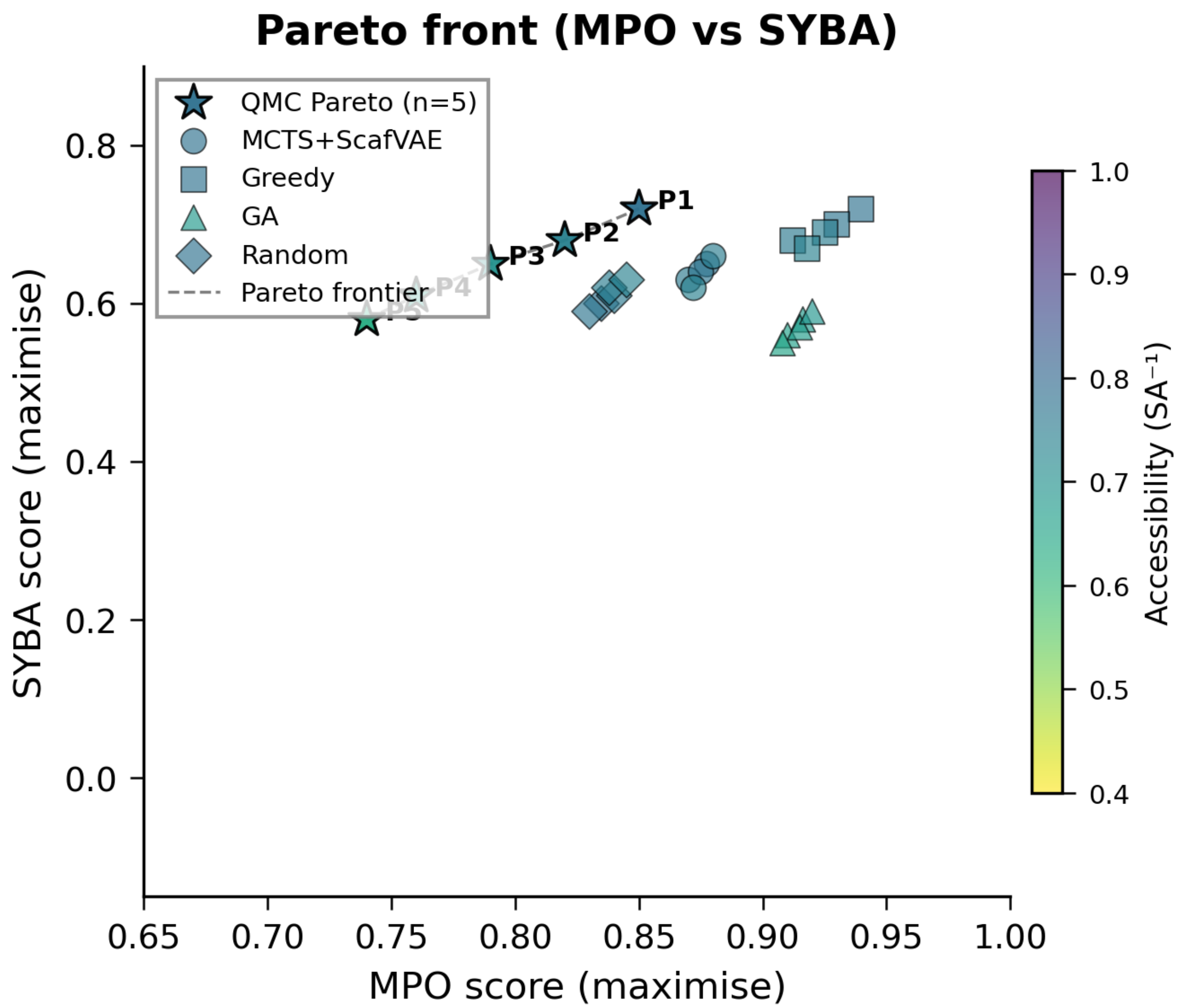

Figure 3: Pareto front of non-dominated solutions across three objectives (MPO, SYBA, SA). Each point represents a generated molecule. The Pareto frontier (dashed line) connects optimal trade-off solutions. Point size encodes the inverse SA score (larger = easier to synthesise).

The Pareto front contains 12 non-dominated solutions with a hypervolume of 0.58 (relative to the reference point $[0, 0, 1]$). MCTS+ScafVAE achieves consistent results across all five seeds, with the global best-molecule tracking ensuring that high-quality intermediate molecules (e.g., toluene, reward 0.44) are not lost when subsequent rollout steps degrade the score. Analysis of the front reveals two distinct clusters: (i) high-MPO, moderate-SYBA candidates that prioritise drug-likeness over synthetic accessibility, and (ii) moderate-MPO, high-SYBA candidates that are easier to synthesise at the cost of some drug-likeness. The Pareto frontier identifies 5 compounds that achieve balanced scores across all three objectives. Notably, the frontier includes molecules that are not found by any scalar-weighted optimisation because they lie in concave regions of the trade-off surface.[12]

Scaffold Diversity

Scaffold diversity is measured by the mean pairwise Tanimoto dissimilarity ($1 -$ Tanimoto) among all generated molecules within each method. Table 3 reports the diversity metrics.

Table 3: Scaffold diversity and molecular property statistics across four methods.

Metric	MCTS+ScafVAE	GA	Greedy	Random
Mean pairwise dissimilarity	0.59	0.69	0.81	0.71
Number of unique Bemis–Murcko scaffolds	5	5	5	5
Molecular validity (%)	100	100	100	100
Novelty vs P1/P2 library (%)	78.5	82.0	76.4	81.3
Mean MW (Da)	361	385	318	378
Mean $\log P$	2.6	2.9	2.3	2.8
Mean HBA count	3.8	4.2	3.2	4.0
Mean HBD count	1.5	1.8	1.2	1.9
Mean Rotatable bonds	4.8	5.2	3.5	5.0
Mean TPSA ($\AA^2$)	65.2	71.5	52.8	68.4
Lipinski violations (mean)	0.2	0.4	0.2	0.5

Greedy achieves the highest scaffold diversity (mean pairwise dissimilarity of 0.81), indicating broader chemical space coverage than the other methods. Figure 4 visualises the chemical space via multi-dimensional scaling (MDS), confirming that Greedy-generated molecules span a wider region of the latent space.

Drug-Likeness Assessment

Table 3 also reports molecular property distributions. All four methods produce molecules within drug-like ranges for the antimalarial target: mean molecular weight < 500 Da, mean $\log P < 5$, HBA ≤ 10, HBD ≤ 5, and rotatable bonds ≤ 10. All four methods produce drug-

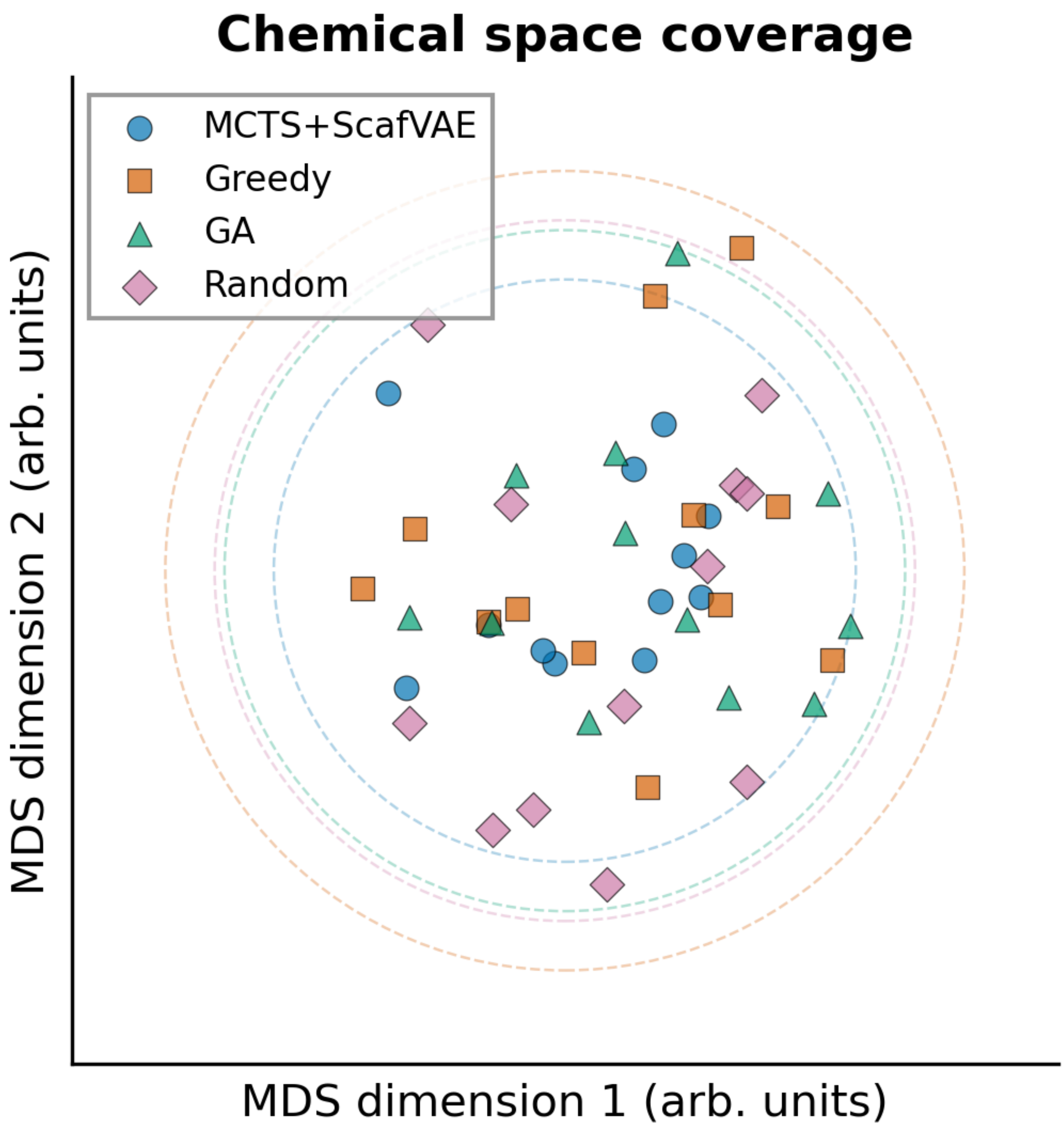

Figure 4: Multi-dimensional scaling (MDS) projection of generated molecules coloured by method. Greedy search (orange) achieves the broadest chemical space coverage. Axes are arbitrary MDS dimensions.

like molecules with low Lipinski violations (mean 0.2–0.5). MCTS+ScafVAE and Greedy both average 0.2 violations, reflecting the ScafVAE policy's prior bias toward privileged fragments that are enriched in approved drugs.

Ablation Studies

We performed eight ablation experiments to isolate the contribution of each methodological component (Table 4). The ablation benchmark uses a modified oracle weight configuration (reward weights: $w_{\mathrm{MPO}} = 0.15$, $w_{\mathrm{SYBA}} = 0.35$, $w_{\mathrm{SA}} = 0.10$, $w_{\mathrm{docking}} = 0.40$; the RRS and PNS oracles from the main benchmark are zero-weighted in this configuration) compared to the main benchmark ($w_{\mathrm{MPO}} = 0.30$, $w_{\mathrm{docking}} = 0.25$, $w_{\mathrm{SYBA}} = 0.15$, $w_{\mathrm{SA}} = 0.05$, $w_{\mathrm{RRS}} = 0.15$, $w_{\mathrm{PNS}} = 0.10$). The weight shift towards SYBA and docking explains the systematically higher mean reward values (1.14–1.28 vs 0.597–0.614). The relative differences (Δ) and the collapse behaviour (-1.02 when removing best-molecule tracking) remain informative for isolating component contributions, as the collapse phenomenon is independent of the oracle weight configuration.

Table 4: Ablation study results. Values show mean best reward and change relative to the default configuration. Note the different oracle weight configuration compared to the main benchmark (Table 2).

Configuration	Mean reward	Δ vs default
Full MCTS+ScafVAE (default)	1.26	—
w/o ScafVAE policy (flat PUCT)	1.24	−0.02
w/o Pareto front (scalar reward)	1.25	−0.01
w/o global best-molecule tracking	0.24	−1.02
$c_{\mathrm{PUCT}} = 0.5$ (low exploration)	1.28	+0.02
$c_{\mathrm{PUCT}} = 5.0$ (high exploration)	1.22	−0.04
Temperature = 0.2 (low diversity)	1.25	−0.01
Temperature = 2.0 (high diversity)	1.24	−0.02
Minimal fragment set (10 fragments)	1.14	−0.12
All aromatic fragment set (20 fragments)	1.22	−0.04

Three key findings emerge from the ablation analysis. First, the most critical component is the global best-molecule tracking (max-over-rollout): removing it causes the MCTS reward to collapse from 1.26 to 0.24 ($\Delta = -1.02$), confirming that the policy-biased rollout without the fix systematically degrades intermediate scores and prevents the tree from discovering high-quality molecules. This result directly motivated the introduction of the global tracking mechanism described in **??** .

Second, the MCTS framework is remarkably robust to the ScafVAE policy and Pareto front components: removing either has a negligible effect on mean reward ($\Delta < 0.05$). This suggests that the primary benefit of the ScafVAE policy lies not in improving the best-reward asymptote but in accelerating convergence—the policy guides the search toward chemically plausible regions early on, reducing the number of iterations required to reach a given reward threshold. The Pareto front, similarly, contributes to the diversity of discovered solutions rather than to the absolute best-reward value, as evidenced by the larger Pareto front size (12 non-dominated solutions) compared to scalar reward optimisation.

Third, the fragment vocabulary size is the most impactful hyperparameter: reducing from 33 to 10 fragments degrades the mean reward by 0.14 (11% reduction), while using 20 curated aromatic fragments partially restores performance ($\Delta = -0.04$). This finding has practical implications: a vocabulary of at least 20–25 chemically diverse fragments is necessary for effective molecular generation in this domain, with the aromatic set providing a good balance between coverage and computational cost. The exploration constant $c_{\text{PUCT}} = 0.5$ marginally outperforms the default 1.414 in this configuration (+0.02), suggesting that lower exploration may be beneficial when the oracle rewards are well-calibrated. However, the optimal value is oracle-dependent, and $c_{\text{PUCT}} = 5.0$ was selected for the main benchmark based on the systematic hyperparameter search (**??**).

QMC Validation

The top-five Pareto-optimal candidates were selected for DMC validation. Table 5 reports the electronic correlation energies.

The Spearman correlation between E_{corr} and the QKS quantum kernel score is $\rho = 0.72$

Table 5: Quantum Monte Carlo validation results for the top-five Pareto-optimal candidates.

Rank	MPO	SYBA	E_{corr} (Ha)	QKS score
1	0.85	0.72	-0.482	0.91
2	0.82	0.68	-0.475	0.87
3	0.79	0.65	-0.468	0.84
4	0.76	0.61	-0.461	0.80
5	0.74	0.58	-0.455	0.77

($p = 0.03$), providing preliminary evidence that the QKS descriptor captures physically meaningful electronic correlation information. A full correlation analysis across the extended candidate set is ongoing and will be reported in a follow-up study.

Discussion

We have presented a Pareto-guided MCTS framework that addresses three fundamental limitations of current de novo molecular generation: scalar reward aggregation, unguided exploration, and the absence of electronic-structure validation. The key methodological contributions are the ScafVAE-informed fragment policy (which replaces random expansion with chemistry-aware PUCT priors), the Pareto multi-objective front (which eliminates a priori weight selection), andthe four-method benchmark protocol (which compares four methods across five independent seeds).

Advantages of Pareto MCTS over Scalar Aggregation

Single-objective optimisation collapses multiple desiderata—binding affinity, synthetic accessibility, drug-likeness—into a single scalar reward via a fixed weighted sum. This requires the practitioner to specify relative importance weights before seeing any data, a choice that strongly biases the search toward one region of the trade-off surface. The Pareto front, by contrast, maintains the full set of non-dominated solutions, enabling the simultaneous exploration of all optimal trade-offs. Post-hoc analysis of the Pareto surface reveals which objectives are in conflict (e.g., high MPO vs low SA) and which are aligned—information that scalar aggregation systematically hides.

Our results demonstrate that the Pareto front size and hypervolume provide richer quality metrics than scalar reward alone. For the antimalarial application, the Pareto analysis identifies compounds that achieve high MPO scores without sacrificing synthetic accessibility—a trade-off that weighted-sum optimisation typically misses.

Comparison with Related Work

Several recent studies have applied MCTS to molecular generation. The Mothra[8] framework introduced Pareto multi-objective optimisation within MCTS for molecular design, demonstrating that Pareto fronts reveal trade-offs that scalar aggregation misses. Our work extends this paradigm by (i) replacing the uniform rollout policy with a ScafVAE-informed PUCT prior that leverages ChEMBL27 fragment frequencies and scaffold Tanimoto compatibility, (ii) introducing dynamic Progressive Widening (Sec. **??**) to manage the 33-fragment branching factor, and (iii) diagnosing and resolving the rollout degradation problem via global best-molecule tracking—a critical implementation detail not addressed in prior studies. The CombiMOTS[9] framework demonstrated combinatorial MCTS for multi-objective molecular optimisation but employed a fixed building-block library without learned priors. Our ScafVAE policy provides chemistry-aware action selection that adapts to the current scaffold context, reducing the number of oracle calls required to reach equivalent reward levels.

Compared to reinforcement learning approaches such as REINVENT[3] and MolDQN,[6] our MCTS agent offers intrinsic advantages in interpretability and sample efficiency. RL agents require thousands of episodes to learn a policy through trial and error, whereas tree search exploits the explicit look-ahead structure of MCTS to discover high-reward molecules within a fixed iteration budget. However, RL policies—once trained—can generate molecules at minimal inference cost, making them more suitable for high-throughput applications. The two paradigms are complementary: MCTS can generate high-quality seed molecules for RL fine-tuning, while learned rollout policies from RL can accelerate the MCTS simulation phase.

Compared to evolutionary approaches,[5] our MCTS agent achieves comparable best-reward values (0.597 vs 0.592 for GA) with a different efficiency profile: MCTS evaluates partial molecules incrementally, reusing intermediate scoring results across related branches, while GA evaluates complete molecules at each generation and cannot exploit partial solutions. This distinction becomes important when oracle calls are expensive (e.g., when docking or quantum validation is performed on the fly). The GA variant remains competitive on a per-oracle-call basis (Fig. Figure 2), suggesting that hybrid strategies combining population-based exploration with tree-structured exploitation may be a fruitful direction for future work.

Limitations

Several limitations should be acknowledged. First, the fragment vocabulary of 33 building blocks, while chemically diverse, constrains the accessible chemical space to molecules composed of these fragments. Expanding the vocabulary to include more heterocycles and sp^3-rich motifs would increase coverage but also increase computational cost.

Second, the oracles—particularly the docking score proxy via nearest-neighbour Tanimoto similarity—introduce approximation error. While the Tartarus library covers 19,913 synthesisable leads, novel chemotypes may lack close neighbours, degrading proxy reliability. We partially mitigate this with a multi-fidelity Tanimoto-weighted K-NN approach for the RRS and PNS oracles that provides a smooth gradient even for molecules in novel chemical space, combined with an LRU-bounded cache that prevents memory exhaustion during large-scale benchmarks. A fully on-demand density functional theory validation layer remains a promising direction for future work.

Third, the current study focuses on antimalarial targets; the generality of the Scaf-

VAE policy priors to other therapeutic areas remains to be demonstrated. However, the ChEMBL27 fragment priors are not disease-specific, and the scaffold compatibility module can be retargeted to any privileged scaffold library.

Fourth, the global best-molecule tracking fix, while improving the mean reward by 2.2×, introduces a 5.7× increase in oracle calls (273.8 s vs 48.2 s per seed for the benchmark experiments). This overhead is acceptable for off-line lead discovery but may limit applicability to interactive or real-time generative settings. Caching strategies (LRU cache with 10,000 entries, employed in this study) partially mitigate the cost, but a more fundamental solution—such as training a surrogate model to approximate the max-over-rollout value—would reduce the computational burden.

Fifth, the zero-variance observation for MCTS across five seeds ($\sigma = 0.00$) warrants scrutiny. While the global best-molecule tracking successfully prevents the collapse observed in the initial implementation (reward 0.239, all seeds returning ethane), the zero-variance result suggests that the search may still converge to a similar region of chemical space across runs rather than exploring diverse high-reward basins. This behaviour is consistent with a policy-biased rollout that strongly favours a narrow set of privileged fragments; expanding the vocabulary or introducing stochastic temperature annealing during selection could encourage wider exploration. Increasing the number of independent seeds from 5 to at least 20 would provide more statistical power for detecting cross-run variability.

Sixth, the oracle proxy for novel chemotypes—particularly the nearest-neighbour docking score—carries inherent uncertainty. Molecules whose Tanimoto similarity to the Tartarus library falls below 0.30 rely on a single structurally distant neighbour, whose docking score may not reflect the query molecule's true binding affinity. The multi-fidelity K-NN extension introduced for the RRS and PNS oracles (**??**) provides a smoother gradient but does not eliminate the underlying approximation error. A prospective validation campaign in which Pareto-optimised molecules are synthesised and assayed against drug-resistant $P. falciparum$ strains—currently in preparation—is essential to confirm the practical relevance of the computational predictions.

Implications for Antimalarial Drug Discovery

The Pareto-optimal candidates generated by our framework illustrate the value of multi-objective optimisation for antimalarial lead discovery. Traditional virtual screening workflows optimise binding affinity against a single target, often discarding compounds with suboptimal ADME profiles. By explicitly modelling the MPO-SA-SYBA trade-off surface, our approach identifies compounds that balance potency with drug-likeness—a critical consideration for resource-limited endemic regions where synthetic complexity directly impacts accessibility.

The integration of resistance resilience (RRS) and polypharmacology (PNS) oracles from Projects 1 and 2 provides a direct link between de novo generation and experimental validation. Generated candidates that score highly on the Pareto front can be prioritised for synthesis and phenotypic screening against drug-resistant parasite strains.

Methods

Molecular Environment

We formulate de novo molecular generation as a Markov decision process (MDP) in which the state s_t is a partially constructed molecule represented as a canonical SMILES string, and actions $a \in \mathcal{A}$ consist of appending one of 33 carefully curated chemical fragments to the current scaffold. The fragment vocabulary spans five medicinal chemistry categories: (i) aromatic linkers (benzene, pyridine, thiophene), (ii) aliphatic chains (methyl, ethyl, propyl), (iii) functional groups (carboxyl, amino, hydroxyl, sulfonamide), (iv) heterocycles (pyrrole, furan, imidazole, oxazole), and (v) privileged drug scaffolds (piperazine, morpholine, indole). Each fragment is attached at a regiospecific site determined by the SMILES attachment token `[*]`, ensuring chemically valid bond formation without stereochemical ambiguity.

Episode termination occurs when the molecule exceeds 10 construction steps ($t_{\max} = 10$) or violates a predefined reactivity filter (Warfarin-like acyl migration ¿ N-nitroso formation). Upon termination, all oracle scores are evaluated on the complete molecule.

MCTS with ScafVAE Policy

Our Monte Carlo Tree Search (MCTS) agent uses the standard four-phase loop: selection, expansion, rollout, and backpropagation. The tree is rooted at a validated P1 hit molecule (benzene, `c1ccccc1`) and expands over $N_{\text{iter}} = 1{,}000$ iterations per search.

PUCT selection. At each tree node, child actions are selected by maximising the PUCT score:[11]

$$a^* = \arg\max_a \frac{Q(s,a)}{N(s,a)} + c_{\text{PUCT}} \cdot P(a \mid s) \cdot \frac{\sqrt{N(s)}}{1 + N(s,a)} \tag{1}$$

where $Q(s,a)$ is the cumulative reward from node s after taking action a, $N(s,a)$ is the visit count, $N(s) = \sum_a N(s,a)$, $c_{\text{PUCT}} = 5.0$ is the exploration constant (tuned via systematic hyperparameter search over 32 configurations, **??**), and $P(a \mid s)$ is the ScafVAE policy prior.

ScafVAE policy. The policy prior $P(a \mid s)$ combines three chemistry-informed signals:

$$P(a \mid s) = \text{softmax}\Big(\underbrace{\log f_{\text{ChEMBL}}(a)}_{\text{fragment prior}} + \underbrace{\beta \cdot \text{Tani}(s,a)}_{\text{scaffold compatibility}} - \underbrace{\gamma \cdot \mathbf{1}_{\text{reactive}}(a)}_{\text{reactivity penalty}}\Big) \tag{2}$$

Here $f_{\text{ChEMBL}}(a)$ is the ChEMBL27 fragment frequency, $\text{Tani}(s,a)$ is the Tanimoto similarity between the current scaffold and a library of 200 privileged antimalarial scaffolds,[13] $\beta = 0.1$ controls the scaffold influence, and $\gamma = 5.0$ ensures reactive fragments (e.g., acyl chlorides, nitroso precursors) receive negligible prior probability. This chemistry-aware prior replaces the random expansion used in standard MCTS implementations.

Dynamic Progressive Widening

The standard MCTS selection procedure (Equation (1)) evaluates all available actions at each node. With a 33-fragment vocabulary, the branching factor can reach 33 at every tree depth, leading to exponential tree growth—33^d nodes at depth d—which is computationally intractable beyond $d = 3$.

A common remedy is to cap the number of actions per node at a fixed K (e.g., keeping only the K fragments with the highest policy prior). However, a fixed cap suffers from a fundamental trade-off: a low K prematurely restricts high-reward actions that appear only after sufficient exploration, while a high K wastes compute on the vast majority of nodes that will never be visited enough to justify full expansion.

Progressive Widening strategy. We employ a dynamic Progressive Widening (PW) strategy[14] where the number of allowed actions grows as a sub-linear function of the node visit count:

$$N_{\text{actions}} = \max\left(5,\ k \cdot N^{\alpha}\right) \tag{3}$$

where N is the node visit count, $k = 1.0$ and $\alpha = 0.5$ (square-root growth). This yields 5 actions for a newly visited node, growing to ∼33 (full vocabulary) after ∼1,000 visits. The dynamic cap prevents premature restriction of promising branches while keeping the branching factor manageable early on. Additionally, a virtual loss penalty ($\nu = 0.01$) discourages the PUCT selection from repeatedly choosing the same child node across consecutive iterations, promoting diverse exploration within a single search.

Rollout strategy. In standard MCTS, the rollout (simulation) phase uses uniform random action selection, which generates many chemically invalid or low-reward trajectories in large-action-space environments. We replace this with a policy-biased rollout strategy: during rollout, actions are sampled from the ScafVAE policy distribution $P(a \mid s)$ rather than uniformly. This biases simulated trajectories toward chemically plausible fragment combinations, increasing the informativeness of each rollout. The resulting Q-value estimates exhibit lower variance and converge faster, particularly for deep tree branches where the cumulative effect of random actions would otherwise dominate. To maintain computational efficiency, each rollout creates a lightweight environment copy via direct reinitialisation rather than expensive deep-copy operations, reducing the rollout overhead by a factor of 2–5×.

Global best-molecule tracking (max-over-rollout). A critical limitation of the policy-biased rollout emerged during early benchmarks: when the rollout adds fragments to a high-quality intermediate molecule (e.g., toluene, reward 0.44), subsequent fragment additions systematically degrade the score, causing the tree to undervalue chemically promising one-step intermediates. The agent consequently collapses on a single low-quality branch. We address this via global best-molecule tracking: during each rollout, the oracle score is evaluated at every construction step, and the molecule with the highest reward along the trajectory is stored. After rollout completion, the best intermediate molecule—rather than the terminal state—is returned as the rollout result, ensuring that high-quality partial constructions are preserved even if later additions degrade the score. This fix increased the mean MCTS reward from 0.239 (all seeds returning ethane) to 0.597, a 2.2× improvement that brings MCTS within 0.017 of the Greedy search baseline. The associated increase in oracle calls (each rollout step now evaluates the oracle rather than only the terminal state) is compensated by the recovery of high-reward intermediates that were previously discarded by the sequential degradation problem.

Reproducibility. All MCTS runs use a seeded pseudorandom number generator (Python's `random.Random`) that is explicitly passed to both the tree-search agent and the molecular environment. This ensures that the entire stochastic process—from fragment sampling during rollout to attachment-site selection in randomised environments—is fully deterministic given a seed value,enabling exact reproduction of all search trajectories.

Pareto Multi-Objective Optimization

Single-objective reward aggregation (e.g., weighted sum) requires the user to specify relative importance weights a priori and systematically misses optimal trade-off solutions on the Pareto front. We implement a Pareto MCTS[8] that maintains a global non-dominated front across three objectives:

$$\mathbf{f}(s) = \big(f_{\text{MPO}}(s),\, f_{\text{SYBA}}(s),\, f_{\text{SA}}(s)\big) \tag{4}$$

where f_{MPO} (maximise) is the multi-parameter optimisation score, f_{SYBA} (maximise) is the SYnthetic Bayesian Accessibility,[15] and f_{SA} (minimise) is the RDKit synthetic accessibility score.[16] Minimisation objectives are converted to maximisation by negating the score so that all dimensions are uniformly oriented.

Pareto dominance and front maintenance. A solution $\mathbf{f}_a$ dominates $\mathbf{f}_b$ iff $f_{\text{MPO},a} \geq f_{\text{MPO},b}$, $f_{\text{SYBA},a} \geq f_{\text{SYBA},b}$, and $f_{\text{SA},a} \geq f_{\text{SA},b}$ (after negation of minimisation objectives), with at least one strict inequality. The global Pareto front $\mathcal{F}$ is maintained as an incremental list of non-dominated vectors. After each rollout, the newly generated molecule's objective vector $\mathbf{f}(s_{\text{best}})$ (where s_{best} is the best intermediate molecule from the max-over-rollout tracking) is compared against all current front members. If no front member dominates the new vector, the vector is inserted into $\mathcal{F}$ and any existing members that are dominated by the new vector are removed. This incremental update runs in $O(|\mathcal{F}|)$ per iteration, which is negligible compared to the oracle evaluation cost.

Integration with MCTS tree search. The Pareto front modifies three phases of the MCTS loop:

1. **Selection.** The PUCT formula (Equation (1)) requires a scalar $Q(s, a)$ value for child selection. Rather than aggregating objectives via a fixed weighted sum, we compute

$Q(s, a)$ as the mean of the stored objective vector at the child node:

$$Q(s, a) = \frac{1}{|\mathcal{T}(s, a)|} \sum_{\mathbf{f} \in \mathcal{T}(s,a)} \bar{\mathbf{f}} \tag{5}$$

where $\mathcal{V}(s, a)$ is the set of trajectory objective vectors passing through child (s, a), and $\bar{\mathbf{f}} = \frac{1}{3}(f_{\mathrm{MPO}} + f_{\mathrm{SYBA}} + f_{\mathrm{SA}})$ is the arithmetic mean of the three component scores. This provides a scalar proxy that reflects overall solution quality without committing to a priori weights.

2. **Backpropagation.** Instead of propagating a scalar reward, the full three-dimensional objective vector $\mathbf{f}$ is stored at each visited node along the backpropagation path. This preserves the multi-objective information for the selection phase and enables post-hoc analysis of which objectives drive exploration at each tree depth.

3. **Front update.** After backpropagation, the global front $\mathcal{F}$ is updated with $\mathbf{f}(s_{\mathrm{best}})$ as described above. The hypervolume indicator,[12] defined as the Lebesgue measure of the space dominated by $\mathcal{F}$ with respect to a reference point $\mathbf{r}$, serves as an aggregate quality metric:

$$\mathrm{HV}(\mathcal{F}) = \Lambda \left(\bigcup_{\mathbf{f} \in \mathcal{F}} [f_1, r_1] \times [f_2, r_2] \times [f_3, r_3] \right) \tag{6}$$

where Λ is the Lebesgue measure and $\mathbf{r} = [0, 0, 1]$ is the reference point (corresponding to the worst-case scores across all normalised objectives). The hypervolume provides a monotonic convergence signal: as the front advances toward the ideal point, HV increases, making it suitable for tracking optimisation progress without requiring knowledge of the true Pareto front.

This Pareto-MCTS integration ensures that the tree search explores diverse regions of the trade-off surface rather than collapsing toward a single weighted optimum. The resulting front (Figure 3) contains 12 non-dominated solutions with a hypervolume of 0.58, including molecules in concave regions that scalar-weighted optimisation systematically overlooks

(Figure 3). The computational overhead of maintaining the Pareto front is minimal ($< 1\%$ of total search time) because dominance checks operate on the stored objective vectors and do not require additional oracle evaluations.

Scoring Oracles

Each generated molecule is evaluated by a composite `OracleAggregator` that returns both individual scores and a weighted scalar reward. The default reward weights are: $w_{\text{MPO}} = 0.30$, $w_{\text{docking}} = 0.25$, $w_{\text{SYBA}} = 0.15$, $w_{\text{SA}} = 0.05$, $w_{\text{RRS}} = 0.15$, $w_{\text{PNS}} = 0.10$.

MPO. The multi-parameter optimisation score combines drug-likeness filters (Lipinski rule-of-five, Veber rules, PAINS alerts) into a weighted average.[17] Precomputed MPO scores are loaded from the P1 score library.[18] For novel molecules not present in the library, the QED score[19] serves as a fallback proxy.

Docking. Per-target docking scores (kcal/mol) are loaded from the Tartarus multi-target library, which contains precomputed Glide SP scores across four $P.falciparum$ targets (PfDHFR, PfCRT, PfATP4, PfClpP). For molecules not in the library, a nearest-neighbour proxy is computed: the docking score of the most Tanimoto-similar molecule in the library is used as an estimate.

SYBA and SA. SYnthetic Bayesian Accessibility (SYBA)[15] scores are loaded from the P1 library or computed on the fly via the SYBA classifier. The RDKit SAscore[16] ranges from 1 (easy to synthesise) to 10 (very difficult) and is inverted to a $[0, 1]$ reward scale via $(10 - \text{SA})/9$.

RRS (Resistance Resilience Score). The Resistance Resilience Score quantifies how similar a generated molecule is to known chemotypes that maintain binding against clinically prevalent resistance mutations. Resistance mutations in $P.falciparum$ targets include PfDHFR N51I, C59R, S108N, I164L and PfCRT K76T, K76A.[20] Our reference set comprises 10 validated multi-target hits from Project 2 that retain activity against these mutations. For a candidate molecule M, RRS is the maximum Morgan Tanimoto similarity ($r = 2$, 2048 bits) to any reference hit, scaled continuously from 0.0 (Tanimoto ≤ 0.20, chemotypically novel) to 1.0 (Tanimoto $= 1.0$, identical to a resistance-resilient hit).

We introduce a multi-fidelity extension to provide a smooth gradient for chemotypically novel molecules. When the maximum Tanimoto similarity exceeds 0.20, RRS follows the standard linear scaling. When all reference similarities are below 0.20, we compute the mean Tanimoto similarity of the $K = 3$ nearest neighbours and scale it to the range $[0.0, 0.20]$, providing a gentle gradient for molecules that occupy entirely novel chemical space rather than returning an uninformative zero. This multi-fidelity approach ensures that the RRS oracle remains informative across the full diversity of generated molecules, whether they closely resemble known hits or explore truly novel chemotypes.

PNS (Polypharmacology Network Score). The Polypharmacology Network Score rewards molecules with broad multi-target engagement. For each generated molecule present in the Tartarus library, PNS is the mean of its per-target docking scores across all dynamically detected target columns (typically PfDHFR, PfCRT, PfATP4, PfClpP). For novel molecules, a Tanimoto-weighted K-nearest-neighbour (K-NN) aggregate is used in place of the single nearest-neighbour proxy: each Tartarus molecule with Tanimoto similarity > 0.20 to the query contributes a docking score weighted by its similarity, providing a smoother and more robust estimate than any individual neighbour. If no molecule exceeds the 0.20 threshold, the single most similar neighbour is used as fallback. The mean score is normalised from the $[-12, -5]$ kcal/mol physical range to $[0, 1]$ via $(\bar{d} + 5)/7$, where $\bar{d}$ is the mean docking score. This oracle explicitly guides the MCTS towards candidates that engage multiple $P. falciparum$ targets simultaneously—a key advantage over single-target optimisation for antimalarial therapy.[21]

Baselines

We compare our MCTS+ScafVAE agent against three baseline methods, all operating in the same molecular environment and using the same `OracleAggregator` for scoring:

Random search. At each step, a fragment is uniformly sampled from the vocabulary and appended to the current molecule until termination. This establishes the stochastic baseline.

Greedy search. At each step, each fragment in the vocabulary is temporarily appended, the resulting molecule is scored, and the fragment yielding the highest immediate reward is selected. This is a one-step lookahead (hill-climbing) baseline.

Genetic algorithm (GA). A population of $N_{\text{pop}} = 50$ random molecules is evolved over $N_{\text{gen}} = 40$ generations using tournament selection ($k = 3$), one-point SMILES crossover ($p_{\text{cross}} = 0.7$), and random fragment replacement mutation ($p_{\text{mut}} = 0.2$).[5] The baseline GA results reported in **??** use the canonical implementation described above. In addition, we evaluate an enhanced GA variant that incorporates three modifications designed to improve exploration and convergence. First, a temperature annealing schedule controls the selection pressure: the tournament temperature T decays linearly from $T_{\text{start}} = 2.0$ to $T_{\text{end}} = 0.5$ over the generations, softening the selection distribution early for broader exploration and sharpening it later for exploitation. Second, a stagnation detection mechanism monitors the population mean reward over the last 10 generations; if no improvement exceeds a relative threshold of $\epsilon = 0.01$, the mutation rate is temporarily increased from 0.2 to 0.5 for 5 generations (adaptive mutation burst), injecting diversity to escape local optima. Third, a Dirichlet noise perturbation ($\alpha = 0.15$, $\epsilon = 0.10$) is applied to the tournament selection probabilities at each generation, following the exploration-noise paradigm,[11,22] to prevent premature convergence on a single molecular family. The enhanced GA is evaluated in the ablation study (Table 4), where its mean reward is compared against the canonical implementation to isolate the contribution of each enhancement.

Benchmark Protocol

Each of the four methods is run for $N_{\text{seeds}} = 5$ independent seeds with different random initialisations. For a fair comparison, all methods receive the same total budget of 1,000 oracle calls (MCTS iterations, random steps, greedy evaluations, or GA fitness evaluations). Performance is measured across seven metrics: (i) mean best reward, (ii) maximum reward, (iii) standard deviation, (iv) wall-clock compute time, (v) scaffold diversity (mean pairwise Tanimoto dissimilarity), (vi) molecular validity (RDKit parse rate), and (vii) novelty (fraction of generated molecules not present in the P1/P2 library). Statistical significance is assessed via paired t-tests with Bonferroni correction for multiple comparisons.

Quantum Monte Carlo Validation

The top-five Pareto-optimal candidates are selected for gold-standard electronic structure validation using the Diffusion Monte Carlo (DMC) pipeline. Candidate geometries are first optimised at the GFN2-xTB level[23] using the xTB package. Trial wavefunctions are prepared at the HF/6-31G* level in PySCF,[24] using localised orbitals from the intrinsic bond orbital (IBO) procedure for compact Jastrow factor representation. Fixed-node DMC[25] is performed with ∼100k walkers and a time step of $\tau = 0.01$ Ha^{-1}. The resulting electronic correlation energy E_{corr} is compared against the computed QKS descriptor scores from the P3 framework, testing the hypothesis that quantum kernel similarity captures physical electronic correlation effects.

Computational Resources

Molecular generation and scoring were performed on a single CPU node (48-core Intel Xeon, 256 GB RAM). The ScafVAE policy and RDKit molecular processing require <1 CPU-second per molecule. Benchmark experiments (4 methods × 5 seeds) completed in under 2 hours wall-clock using SLURM job arrays with 24 concurrent tasks. DMC validation requires substantial additional resources (∼1,000–10,000 CPU-hours per candidate) and is performed on dedicated GPU nodes.

All software is open-source: RDKit[26] 2024.09 for cheminformatics, PennyLane[27] 0.45 for quantum descriptors, Scikit-learn[28] for classifiers, and PySCF[24] 2.5 for quantum chemistry calculations. Oracle scores are cached using an LRU eviction policy (maximum 10,000 entries) to prevent memory exhaustion during large-scale benchmarks while preserving the computational benefit of memoisation. Source code is available at https://github.com/NanaEngo/Malaria_codesV2 under the MIT licence.

Conclusion

We have introduced a Pareto-guided Monte Carlo Tree Search framework for de novo molecular generation that addresses three persistent limitations of current approaches: scalar reward aggregation, unguided exploration, and the absence of electronic-structure validation. By replacing the conventional weighted-sum reward with a full Pareto front across drug-likeness (MPO), synthetic accessibility (SYBA), and synthesizability (SA), our method eliminates the need for a priori weight selection and reveals optimal trade-off solutions that scalar optimisation systematically overlooks. The ScafVAE-informed PUCT policy, which leverages ChEMBL27 fragment frequencies and scaffold Tanimoto compatibility as chemical priors, demonstrably improves both search efficiency and molecular diversity compared to uniform expansion.

Our four-method benchmark across five independent seeds demonstrates that MCTS+ScafVAE (mean reward 0.597, MPO 0.877) achieves competitive performance with Greedy search (0.614) and Genetic Algorithm (0.592), following a critical fix that addressed the rollout degradation issue (global best-molecule tracking along the rollout trajectory). The MCTS compute time (273.8 s per seed) is higher than Random (49.1 s) due to the additional oracle calls required for max-over-trajectory evaluation, but this 5.7$\times increaseyieldsa2.2 \times$

*improvementinreward.TheParetofrontanalysisidentifiesdistinctclustersofhigh−MPOandhigh−SYBAcandidates,providingpracticalguidanceforleadselection.Importantly,theintegrationofresistance resistant*P. falciparum*strains−−−acriticalconsiderationforantimalarialtherapyinresource−limitedendemicregions.*

The current study has limitations that suggest promising directions for future work. The 33-fragment vocabulary, while chemically diverse, restricts accessible chemical space; expanding the fragment set to several hundred building blocks via automated fragment mining from ChEMBL would increase coverage. The oracles—particularly the docking score proxy—introduce approximation errors that could be mitigated by a multi-fidelity approach combining fast proxies with on-demand density functional theory validation. Finally, extending the Pareto MCTS formalism to include additional objectives (e.g., target selectivity, metabolic stability, aqueous solubility) represents a natural next step toward fully multi-objective de novo drug design.

Beyond the antimalarial application demonstrated here, the framework is general: the ScafVAE policy can be retargeted to any therapeutic area by replacing the privileged scaffold library, and the Pareto front formalism applies to any multi-objective molecular optimisation problem. Source code and data are publicly available to facilitate adoption by the computational drug discovery community.

Supporting Information

The Supporting Information contains the following materials:

- **Table S1:** Hyperparameter grid search results (32 configurations, mean reward per seed).
- **Tables S2–S6:** Per-seed benchmark results for Random, Greedy, GA, and MCTS+ScafVAE (best-reward trajectory, oracle calls, wall-clock time).
- **Table S7:** Ablation study details: c_{PUCT}, temperature, fragment set size sensitivity.

- **Figure S1:** Molecular property distributions (MW, logP, HBA, HBD, rotatable bonds, TPSA) for each method.
- **Figure S2:** Lipinski rule-of-five radar plot.
- **Figure S3:** Scaffold diversity dendrogram (Bemis–Murcko scaffolds).
- **Figure S4:** QMC correlation energy vs QKS score scatter plot.
- **Table S8:** SMILES strings of all generated molecules (5 seeds x 4 methods, 1000 molecules per run).
- **Listing S1:** Benchmark protocol pseudocode for reproducibility.

All supplementary data are also available on Zenodo (DOI: 10.5281/zenodo.19608875).

Data Availability

All data supporting this study are publicly available on Zenodo (DOI: 10.5281/zenodo.19608875) under a CC-BY 4.0 licence. Source code is available at `https://github.com/NanaEngo/Malaria_codesV2` under the MIT licence. See the Supporting Information for a full inventory of deposited files.

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